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ROAST MY DESIGN DECISIONS

Real-life trade-offs, success and failure

SOFICO
SOFTWARE DRIVING MOBILITY

Steven Hillaert – solution architect

Live | Multiple Choice | 0 votes | Stop

What is your role?

Waiting for responses...

Results will appear here as people respond


Don’t follow my decision blindly
These decisions were made by balancing tradeoffs in a specific context:
Functional
Non-functional
Technical
Strategic

Use-case 1 – the bank

System to calculate commissions

- Account holders
- Accounts
- Transactions
- Documents
- Agents

This impacts someone's income!

The bank

I designed a relational database without foreign keys.



Closed | Rating Scale | 0 votes | Start

The bank - poll 1

Do you trust my database?

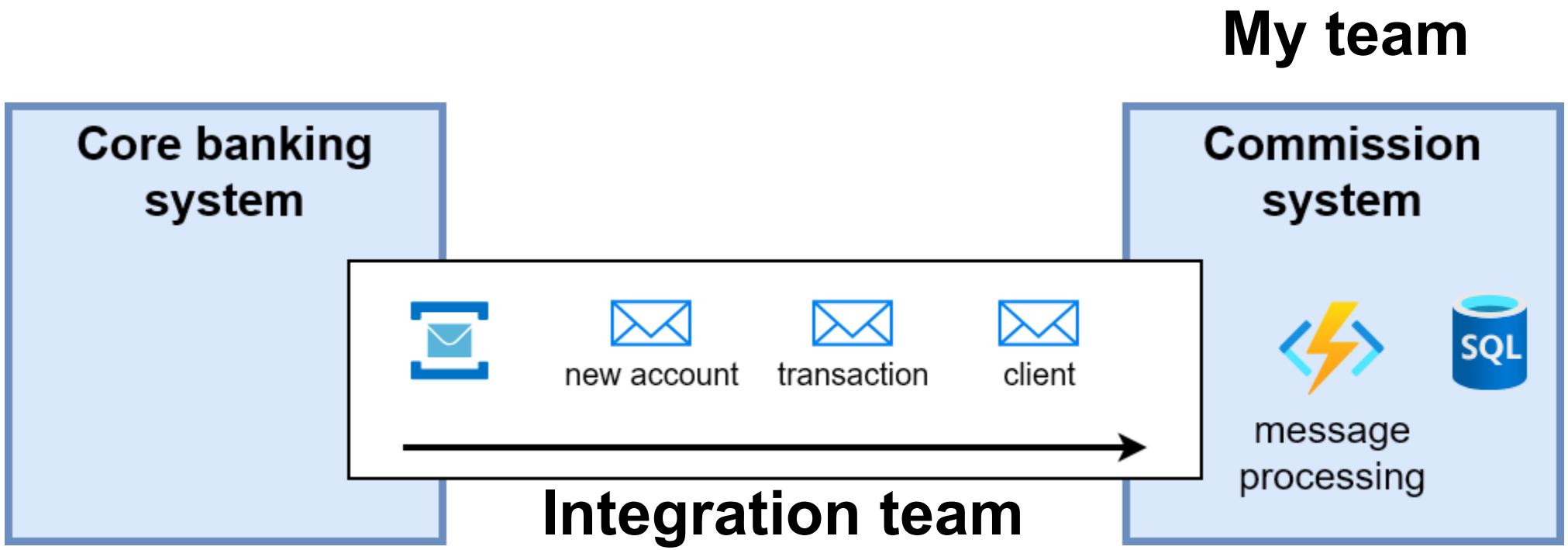
No responses received

This poll was closed without receiving any responses

The bank – project constraints

- Built as consulting service
- We deliver the application
- We only do 3rd level support
- Fixed time and budget
- Validate latest messages are processed

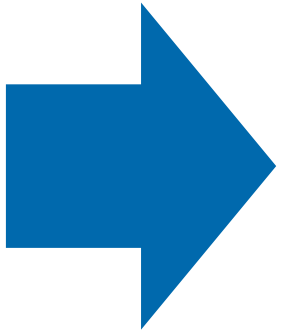
The bank – system



The bank – standard challenges ⚠️

Standard challenges when using message brokers

- At-least-once delivery
- Out-of-order processing
- Poison messages
- Monitoring



- Inbox pattern
- Idempotency
- Retries
- Built in Azure Service Bus 💶
- Tooling (NServiceBus) 💶

Idempotent messaging will save you from a lot of grief

The bank – integration challenges ⚠️

- Multiple messages for the same event
  - → 2 create messages, identical data, different timestamp
- Different messages for the same event
  - → 1 update and 1 create message, different timestamp, any order
- Out-of-order publishing
  - → Updates or deletes before creates
  - → Transactions before accounts (child entities before parents)

The bank – integration solutions 💡 (with FK)

- Multiple messages for the same event (2 creates)
  - ➔ Keep the one with most recent timestamp
- Different messages for the same event (update + create)
  - ➔ Treat both as upsert, keep the one with most recent timestamp
- Out-of-order publishing
  - Updates or deletes before create ➔ up-del-sert
  - Child entities before parents ➔ Retry until parent exists

The bank – current status

Still using foreign keys

- ✔️ Fixed standard messaging challenges with idempotency
- ✔️ Fixed quirky messages from producer with up-del-sert & idempotency
- ✔️ Avoided extra tooling/complexity
- ✔️ Avoided extra costs

- ❌ Child entity messages before parent can be solved with retries
  - ⚠️ monitoring and manual interventions
  - ⚠️ poison queues (can’t keep retrying)
  - ⚠️ verification with source system is difficult

The bank – why do we use FK's?

FK's enforce referential integrity

The bank – why that breaks the system

Can’t create child entities before parents

Can’t create transaction before account

The bank – do we need it?

Do we need referential integrity?

(for this project)

The bank – do we need it?

Not really!

- We're not owner of that data
- Calculation behaves correctly with missing data
- Eventually consistent
- Messages can be processed as they come (no retries)
- Monitoring inconsistencies is easier from a database

The bank – socio-technical impact

- Integration with source system isn't reliable
- Integration team and my team are external, different suppliers
- The original teams won't be there forever
- Support will be given by people who didn't develop the system
- **The integration shouldn't create friction for the customer**

Closed | Rating Scale | 0 votes | Start

The bank - poll 2

How much trust in the current solution?

No responses received

This poll was closed without receiving any responses

The bank – core lessons

- Take a step back and look at the bigger picture
- Dare to challenge the “defaults”
- Architecture is not about technical purity
- Why not event sourcing?

Use case 2 – the inspection

Goal:

Standardize decision-making concerning damages

(repair, insurance, invoicing…)

Users:

Professionals, fleet managers, drivers

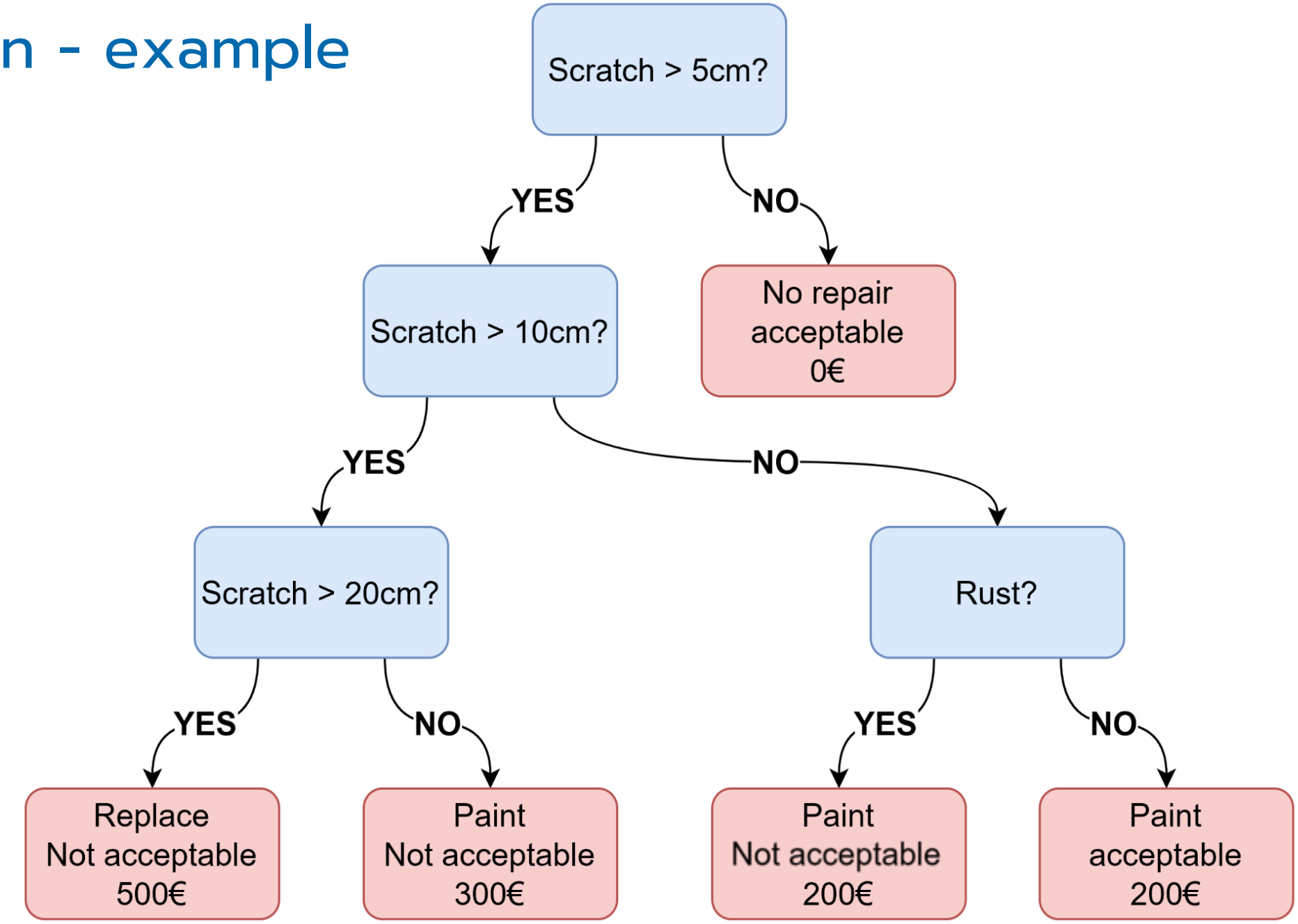
Solution:

Decision tree with questions

(per customer, damage,…)



The inspection - example



The inspection – the simplest solution

A tree structure where every node is a plain text yes/no question.

- Show the text (question) to the user
- User answers yes or no
- Show the next question
- Until you reach the end node

Workload: +/- 2 weeks

That’s not what I did!



The inspection – the not so simple solution

- No dumb text-based questions
- Intelligent properties & comparators & type safe
- Builder & factory patterns
- Visitor pattern (traverse the tree)
- Supporting designer in the UI for configuration
- Multilingual

Workload: +/- 10 weeks

Closed | Rating Scale | 0 votes | Start

The inspection - poll 1

Do you trust my overengineering?

No responses received

This poll was closed without receiving any responses

The inspection – a little philosophy

- As an architect, you're able to see the greater picture
- You're involved with stakeholders at strategic level
- It's not about predicting the future, nor building for **that** future
- It's about supporting a strategy & vision
- And supporting multiple futures
- **Will your decisions support or hinder it?**

The inspection – what drove me

- **Changing** landscape of users
- **Data** is key, pure text doesn't create useful data
- Plain text questions don't **scale** very well
- Decisions need "metadata", everything needs to be **smart**
- **AI**/machine learning was being explored

The inspection – my decision

These things drove me to engineer it properly and completely, from the first version.

Too many future paths would need this. Even outside the inspection scope, as basis for a future rule engine.

We were agile in the capabilities offered but not in the core engineering.

Agile is not about creating the fastest and simplest solution!

Closed | Rating Scale | 0 votes | Start

The inspection - poll 2

You trust my overengineering now?

No responses received

This poll was closed without receiving any responses

The inspection – a few weeks later

"We would like to have the actual values"

"Inspectors are tired of answering 20 questions"

"Would this be compatible with AI?"

Starting with the simplest solution can be a risk for the future. (under-engineering)

We're conditioned to build on what we already have



Use case 3 – the microservices

- Small development team
- 1 monolithic system
- Supporting the core business
- Everything on-premise

- Growing team (x4)
- More customers and users
- Moving to the cloud
- European projects
- Expanding beyond core business
- Cohesion between the new opportunities

The microservices – why?

- **Scalability** must increase to support the load
- **Reuse** must increase for shared data and services
- **Integration** must increase to create synergy
- New applications need more **autonomy**
- Growing teams need more **focus**
- Avoid single huge **deployments**
- We would do it **gradually**, we had a **plan**

Closed Rating Scale 0 votes Start

The microservices - poll 1

Were microservices the right choice?

No responses received

This poll was closed without receiving any responses

The microservices – the struggle

- Started with **foundation** services
- Built **core** business services on top
- Still using a **single database**
- Services started to **communicate** between each other (a lot)
- **Dependencies** got stronger
- Problems were fixed as **isolated** cases (distributed transactions)
- Teams were busy, **no governance**

The microservices – core lessons

- Choosing microservices was a good choice
- The implementation failed, we treated it as a technical problem
- Boundaries & ownership are the real challenges
- Local optimizations can destroy global coherence
  - Focused on get it working
  - Architecture drift
- Reuse is overrated

Use case – adapt skills or architecture

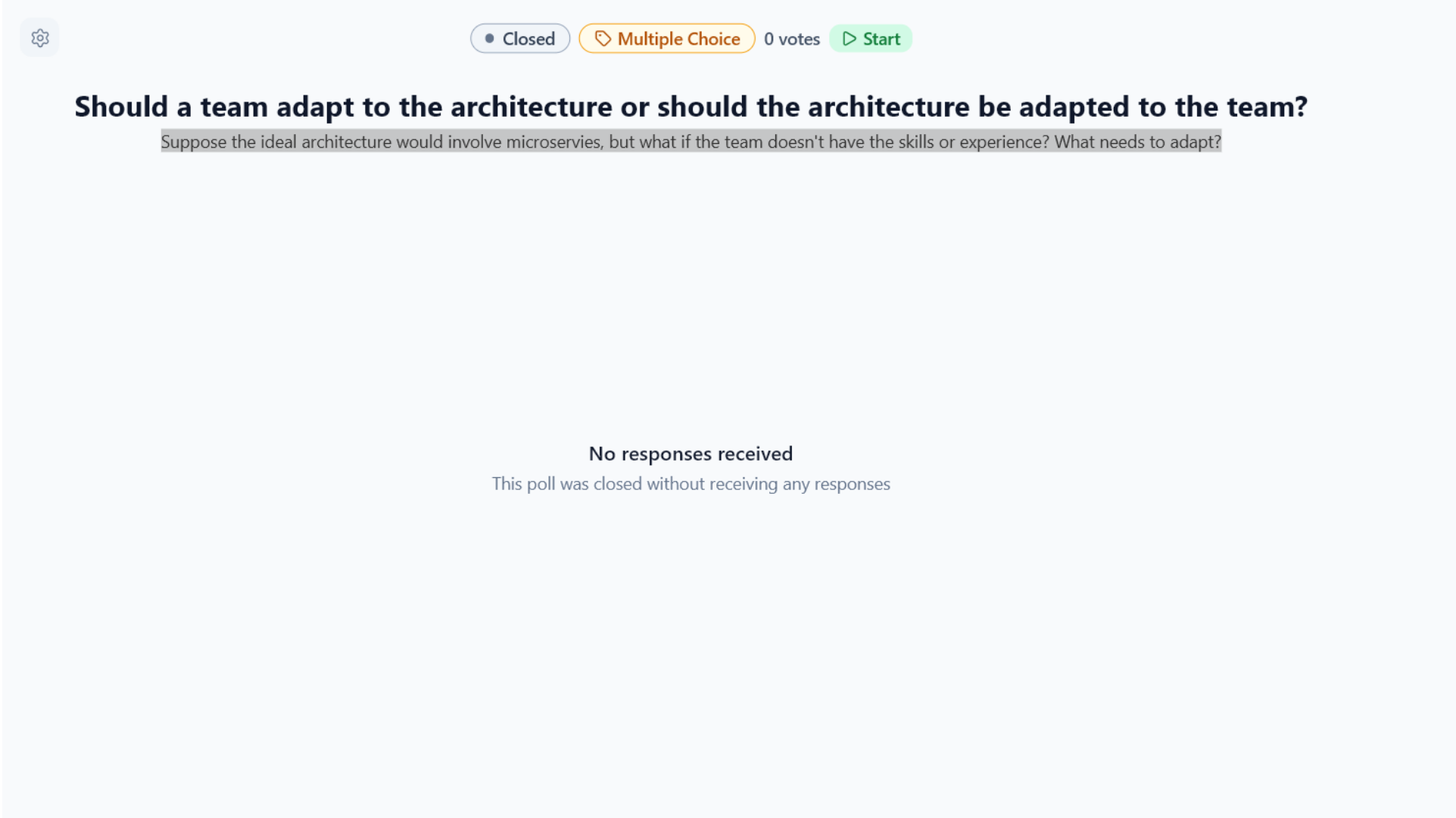
Should a team adapt to the architecture

or

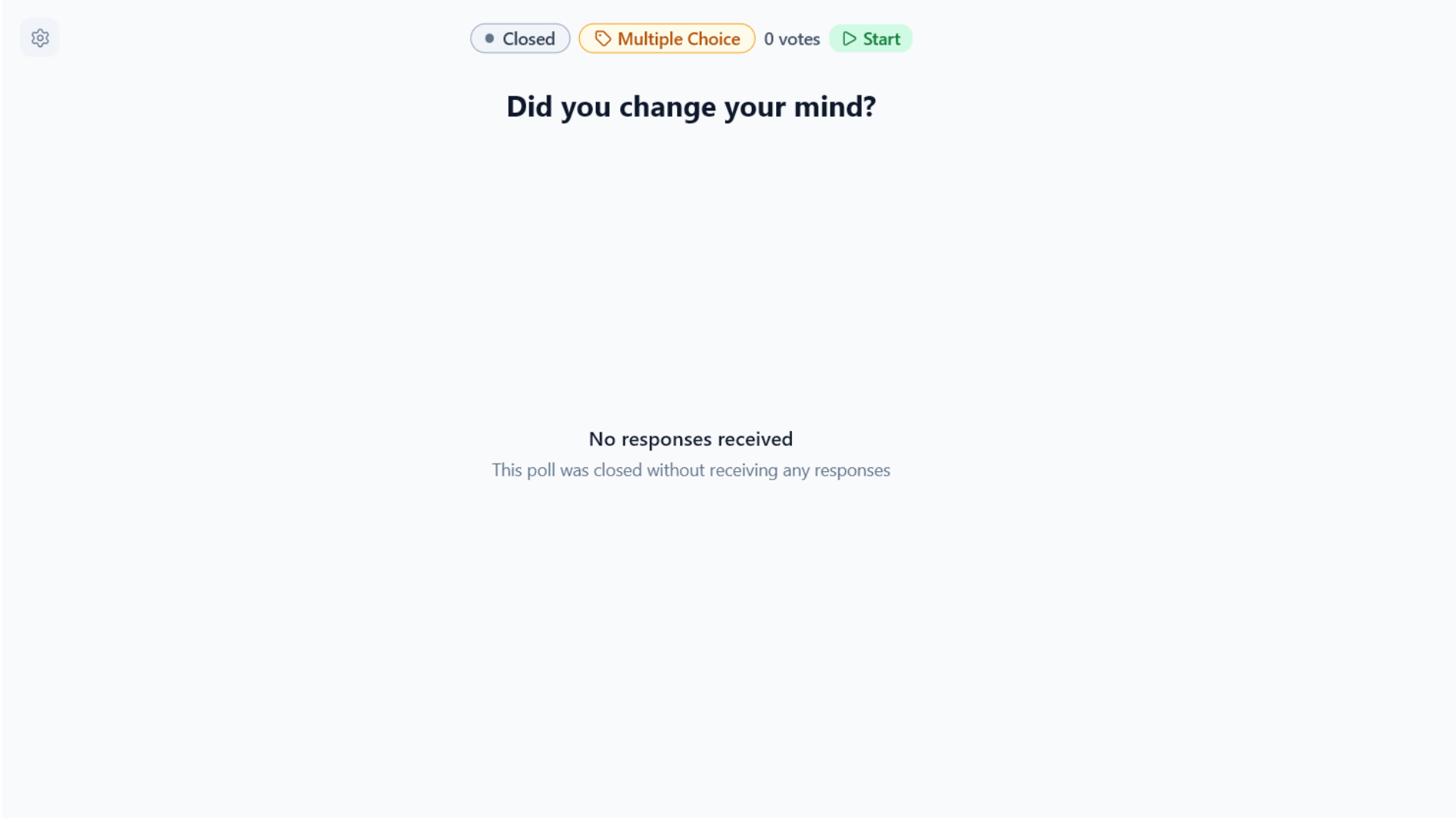
should the architecture be adapted to the team?

Suppose the ideal architecture would involve microservies, but what if the team doesn't have the skills or experience? What needs to adapt?

Use case – adapt skills or architecture



Use case – adapt skills or architecture



The goal of software design and architecture

We write software to solve today's problems.

But we must design and architect solutions
to continue delivering value
in an uncertain, constantly changing future.

Don't just look at upsides of a trade-off.
Can you also live with the downsides?

feedback

Thank you



Questions?